4–12 Find R_{IN} in Figure P4–12.

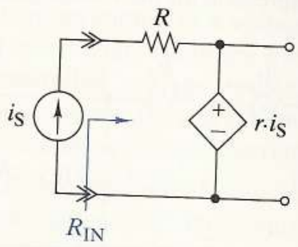


FIGURE P4–12

4–13 Find R_{IN} in Figure P4–13.

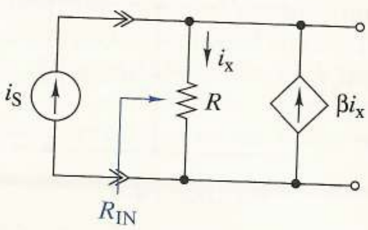


FIGURE P4–13

4–14 Find the Thévenin equivalent circuit seen by the load in Figure P4–14.

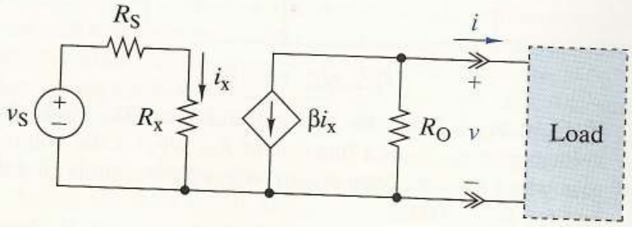


FIGURE P4–14

4–15 Find the Norton equivalent circuit seen by the load in Figure P4–15.

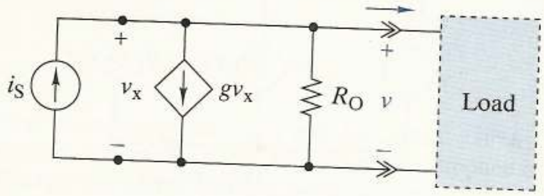


FIGURE P4–15

4–16 The circuit parameters in Figure P4–16 are $R_B =$ $R_C = 4\text{ k}\Omega, \beta = 120, V_\gamma = 0.7\text{V}$, and $V_{CC} = 15\text{ V}$. Fin v_{CE} for $v_S = 2$ V. Repeat for $v_S = 5$ V.

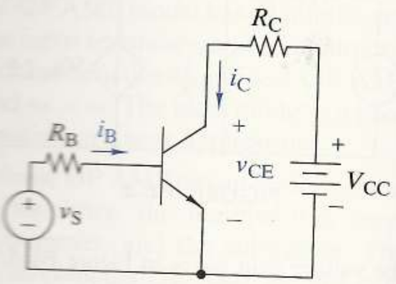


FIGURE P4–16

4–17 **D** The circuit parameters in Figure P4–1 $R_C = 3\text{ k}\Omega, \beta = 100, V_\gamma = 0.7\text{ V}$, and $V_{CC} = 5\text{ V}$. Select ue of R_B such that the transistor is in saturation mode $v_S \geq 2$ V.

4–18 The parameters of the transistor in Figure P4– $\beta = 50$ and $V_\gamma = 0.7$ V. Find i_C and v_{CE} for $v_S = 0.8$ V peat for $v_S = 2$ V.

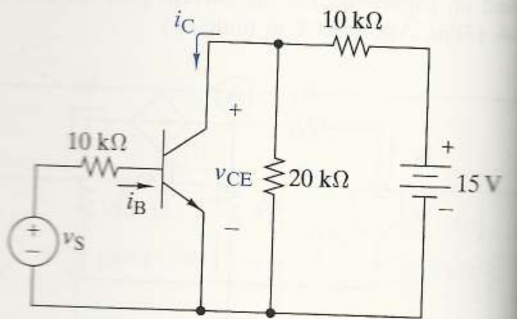


FIGURE P4–18

OBJECTIVE 4–2 OP AMP CIRCUIT ANALYSIS (SECTS. 4–3, 4–4)

Given a circuit consisting of linear resistors, OP AMP independent sources, find selected output signals or output relationships.

See Examples 4–13, 4–14, 4–16, 4–17, 4–18 and Exercises 4–14, 4–15, 4–17, 4–18, 4–19, 4–20, 4–21

4–19 **E** Two OP AMP circuits are shown in Figure Both claim to produce a gain of either +10 or – 10.

(a) Show that the claim is true.

(b) A practical source with a series resistor of 1 connected to the input of each circuit. Does the or claim still hold? If it does not, explain why.

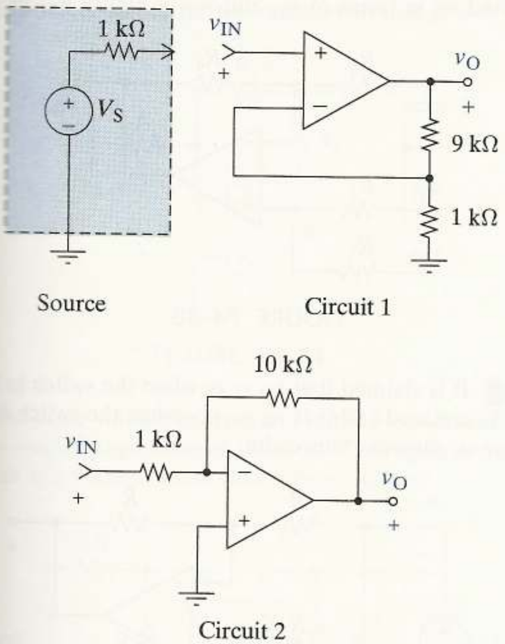


FIGURE P4–19

4–20 E Suppose the output of the practical source shown in Figure P4–19 needs to be amplified by –100 and you can use only the two circuits shown. How would you connect the circuits to achieve this? Explain.

4–21 (a) Find v_O in terms of v_S in Figure P4–21.
(b) Validate your answer by simulating the circuit in OrCAD.

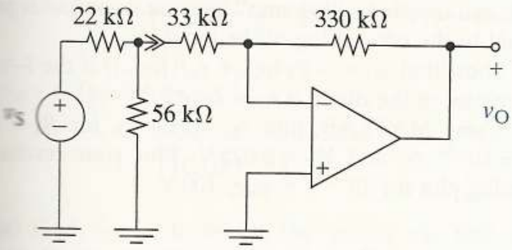


FIGURE P4–21

4–22 What is the range of the gain v_O/v_S in Figure P4–22?

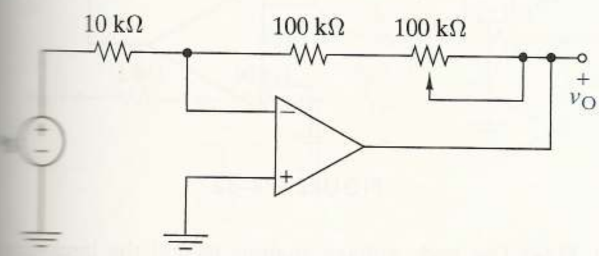


FIGURE P4–22

4–23 **(a)** Find v_O in terms of v_S in Figure P4–23.
(b) Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

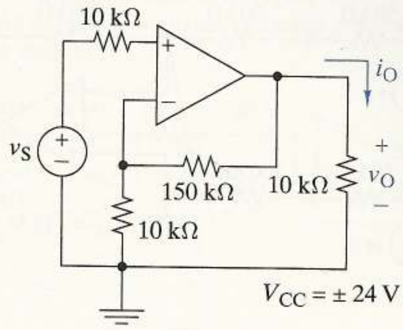


FIGURE P4–23

4–24 **(a)** Find v_O in terms of v_S in Figure P4–24.
(b) Find i_O for $v_S = 2$ V.

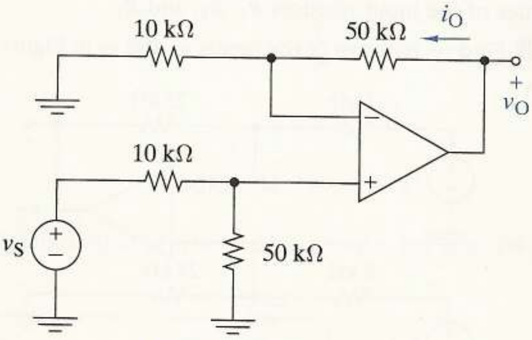


FIGURE P4–24

4–25 What is the gain v_O/v_S in Figure P4–25 when the switch is in position 1? Repeat for positions 2 and 3.

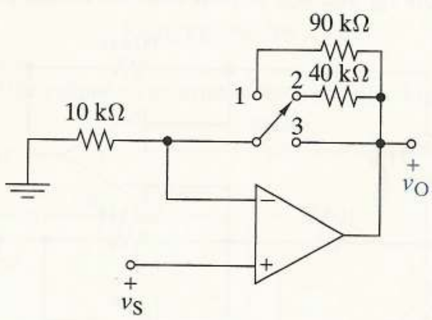


FIGURE P4–25

4–26 **(a)** Find v_O in terms of the inputs v_1 and v_2 in Figure P4–26.

(b) If $v_1 = 1\,\mathrm{V}$, what is the range of values v_2 can have without saturating the OP AMP?

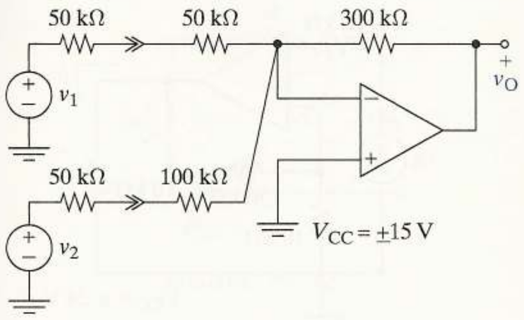


FIGURE P4–26

4–27 The input-output relationship for a three-input inverting summer is

$$v_O = -[v_1 + 10v_2 + 100v_3]$$

The resistance of the feedback resistor is $100\,\mathrm{k\Omega}$. Find the values of the input resistors R_1, R_2, and R_3.

4–28 Find v_O in terms of the inputs v_1 and v_2 in Figure P4–28.

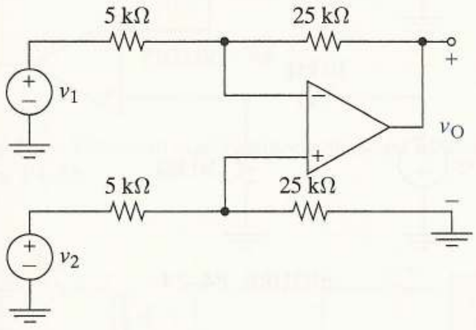


FIGURE P4–28

4–29 The switch in Figure P4–29 is open. Find v_O in terms of the inputs v_{S1} and v_{S2}. Repeat with the switch closed.

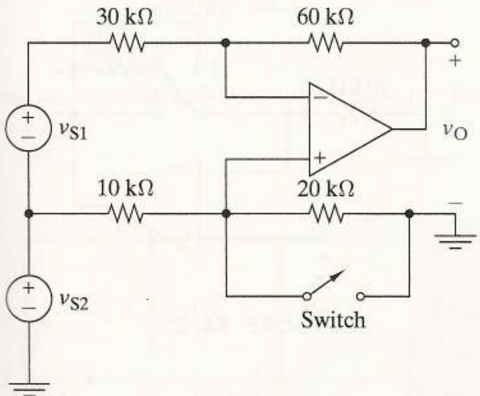


FIGURE P4–29

4–30 Find v_O in terms of v_{S1} and v_{S2} in Figure P4–30.

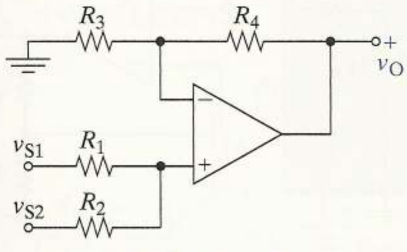


FIGURE P4–30

4–31 E It is claimed that $v_O = v_S$ when the switch in Fi P4–31 is closed and that $v_O = -v_S$ when the switch is o Prove or disprove this claim.

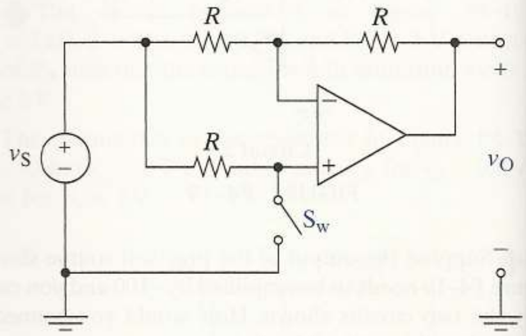


FIGURE P4–31

4–32 The circuit in Figure P4–32 has a diode in its feedb path and is called a "log amp" because its output is pro tional to the natural log of the input.

(a) Show that $v_O = -V_T \ln(1 + v_S/(R_S I_o))$ if the i–v c acteristics of the diode is $i_D = I_O(e^{v_D/V_T} - 1)$.

(b) Using MATLAB, plot v_O versus v_S for $R_S = 10$ $I_O = 10^{-14}$ A, and $V_T = 0.025\,\mathrm{V}$. Plot your results o semilog plot for $10^{-6}\,\mathrm{V} \le v_S \le 100\,\mathrm{V}$.

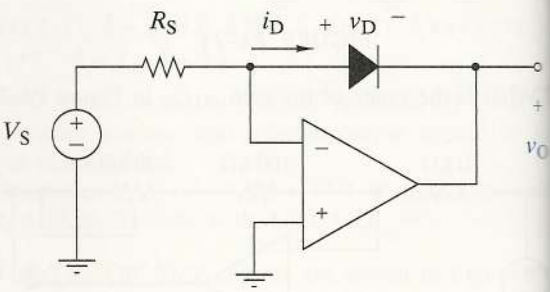


FIGURE P4–32

4–33 **(a)** Use node voltage analysis to find the input-ou relationship or K of the circuit in Figure P4–33.

4–40 Find the output v_O in terms of the input v_1 in Figure P4–40. Verify your answer using OrCAD.

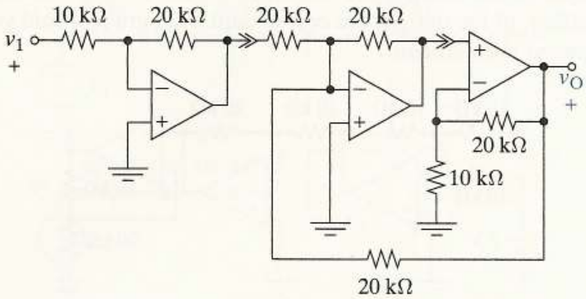


FIGURE P4–40

Objective 4–3 OP AMP Circuit Design and Evaluation (Sect. 4-5)

Given an input-output relationship, use resistors and OP AMPs to design one or more circuits that implement the relationship within the stated constraints. Choose the better of two working circuits to meet certain constraints.
See Design Examples 4–12, 4–15, 4–20, 4–21, 4–22, 4–23, 4–24, Design Exercises 4–12, 4–16, 4–22, 4–28, 4–29, 4–30, Evaluation Examples 4–19, 4–20, 4–21, 4–22

4–41 D Design an OP AMP amplifier with a voltage gain of −10 and an input resistance greater than 10 kΩ using standard 5% resistance values less than 300 kΩ.

4–42 D Design an OP AMP amplifier with a voltage gain of 4 using only 10-kΩ resistors and one OP AMP.

4–43 D Using a single OP AMP, design a circuit with inputs v_1 and v_2 and an output $v_O = 2v_2 - 3v_1$. The input resistance seen by each input should be greater than 5 kΩ.

4–44 D Design a differential amplifier with inputs v_1 and v_2 and an output $v_O = 10(v_2 - v_1)$ using only one OP AMP. All resistances must be between 5 kΩ and 200 kΩ.

4–45 D Using no more than two OP AMPs, design an OP AMP circuit with inputs v_1, v_2, and v_3 and an output $v_O = -3v_1 + 2v_2 - 5v_3$.

4–46 D Design the interface circuit in Figure P4–46 so that 10 mW is delivered to the 100-Ω load. Repeat for a 100-kΩ load. Verify your designs using OrCAD.

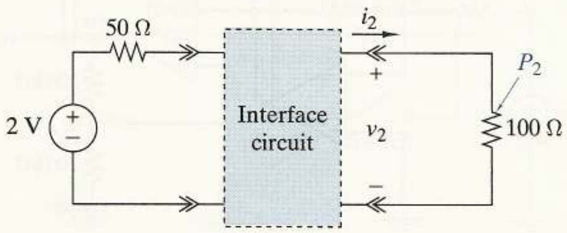


FIGURE P4–46

4–47 D Design the inter... the output is $v_2 = 150v_1$ ...

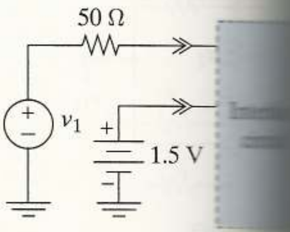


FIG...

4–48 D (a) Design a ci... 5 V using two OP A...
(b) Repeat using on...

4–49 E A requirement ... input-output relatio...

Two proposed designs ... project engineer, you ... for production. Which ... for production and wh... perform the required ...

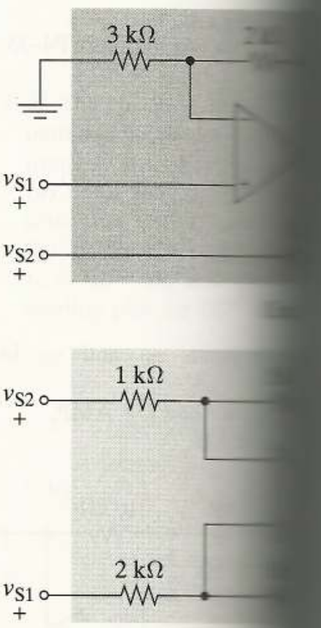
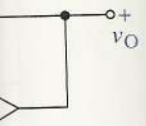


FIG...

4–50 E A requireme... deliver 12 V to a 1-k...

in Figure P4–3



30

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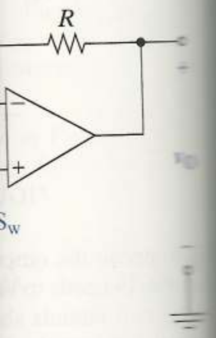
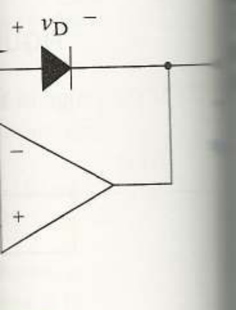
s when the switch



-31

has a diode in its
ecause its output is
input.
$v_S/(R_S I_o)$) if the
$_O(e^{v_D/V_T}-1)$.
versus v_S for R_S =
V. Plot your resu
100 V.



4-32

ysis to find the inpu
t in Figure P4–33.

Select values for the resistors so that $K = -3$.

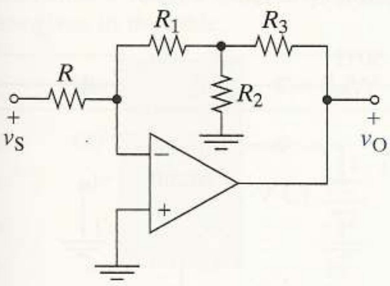


FIGURE P4–33

e node voltage analysis in Figure P4–34 to show that $= -v_S/2R$ regardless of the load. That is, show that the is a voltage-controlled current source.

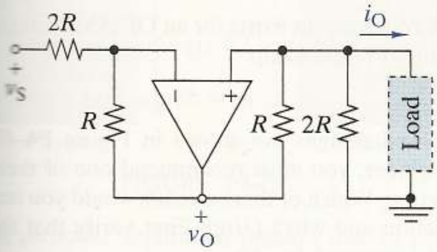


FIGURE P4–34

Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4–35.

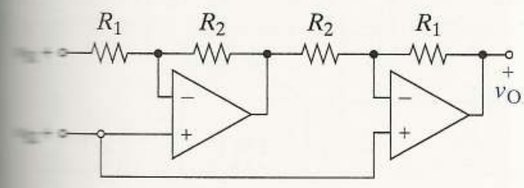


FIGURE P4–35

Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4–36.

Redesign the circuit using only one OP AMP. Validate your design using OrCAD.

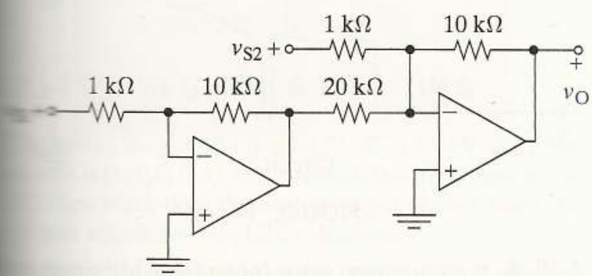


FIGURE P4–36

4–37 E On an exam, students were asked to design an efficient solution for the following relationship: $v_2 = 3v_1 + 15$. Two of the designs are shown in Figure P4–37. Which, if either, of the designs are correct and what grade would you award each student?

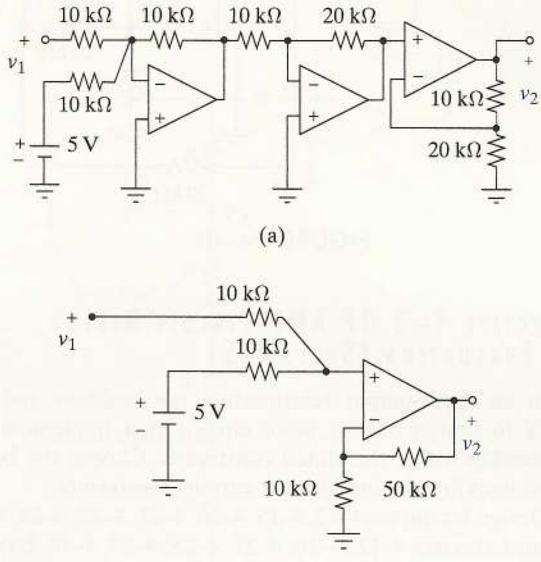


FIGURE P4–37

4–38 Find the output v_2 in terms of the input v_1 in Figure P4–38.

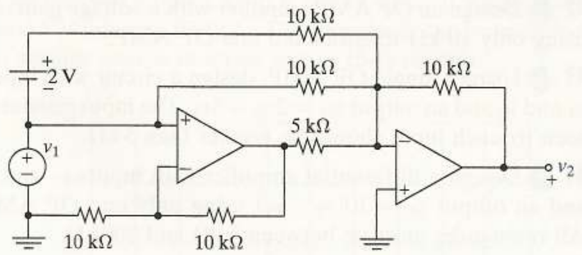


FIGURE P4–38

4–39 Find the output v_O in terms of the input v_S in Figure P4–39.

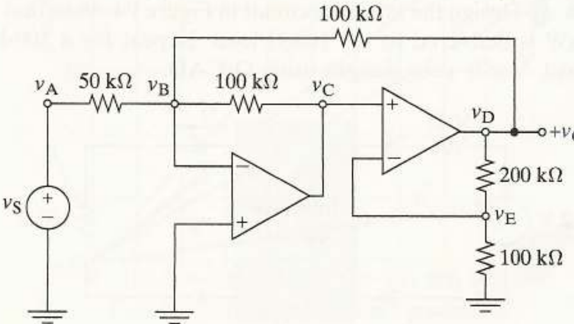


FIGURE P4–39